

Pratham Patel

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EDUCATION

Bachelor of Science (Honours) in Computer Science

(Expected) May 2027

York University

Toronto, ON

- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Database Systems, Software Testing, Data Mining, Web Development, Operating Systems, Computer Vision

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C/C++, Bash, SQL

Libraries & Frameworks: Spring Boot, React.js, Angular, Node.js, FastAPI, Flask, TailwindCSS, Redux, Scikit-learn, TensorFlow, NumPy, Pandas

AI & ML: AWS Bedrock (Claude), LLM Prompt-Chaining, RAG, Agentic AI, Intent Classification, Model Evaluation, LSTM

Cloud & DevOps: AWS, Azure, GCP, Azure DevOps, Docker, Kubernetes, Jenkins, CI/CD Pipelines, Redis, Bitbucket

Tools: Postman, SoapUI, Karate, Selenium, JIRA, Git, GitHub, IntelliJ, VS Code, Guidewire, Power BI, Linux/Bash

EXPERIENCE

Software Engineer Intern, AI & Automation

Jan 2026 – Aug 2026

Aviva Canada

Markham, ON

- Reduced test-authoring time by **80%** by engineering and deploying an end-to-end **AI productivity platform** via **Azure DevOps**. Built on **AWS Bedrock** (Claude) using prompt chaining, JSON schema validation, and context-grounded generation to automatically produce structured API and UI test cases from Jira stories via **REST APIs** across a large-scale production platform.
- Achieved **90% intent-recognition accuracy** by building a Copilot-style conversational AI agent with an **LLM-driven intent-classification router** and a multimodal **RAG context layer** (images, PDFs, documents), enabling non-technical QA teams to refine test suites using natural language.
- Cut auto-insurance quote completion time by **40%** by engineering a full-stack **agentic AI chatbot** (**Angular, TypeScript, Java, Spring Boot**) that automated intake via multi-turn NLP and real-time Guidewire data validation.
- Designed and built an **API test automation framework** for Guidewire PolicyCenter from scratch using **Karate (Java, Gradle)**, **Postman**, and **SoapUI**, architecting data-driven regression suites to validate REST endpoints and data contracts across quote-to-bind insurance workflows.
- Automated **200+ regression test cases** across **distributed Guidewire systems** (PolicyCenter, ClaimCenter, BillingCenter), cutting end-to-end regression cycle time by **80%** and surfacing critical integration defects pre-production via parallel **CI/CD** execution on **Jenkins**.

Software Engineer Intern

May 2025 – Aug 2025

TourWalk – M.U.T Services Pvt. Ltd.

Remote

- Boosted system reliability and data throughput by **30%** by engineering optimized **RESTful APIs** in **Node.js** and implementing a **Redis** caching layer to handle dynamic real-time travel queries at scale.
- Improved booking-flow conversion by **15%** (internal UX metrics) by architecting responsive **React.js** and **Redux** front-end interfaces with optimized state management, reducing drop-off across key conversion steps.
- Achieved **100% security audit pass rate** with zero incidents by integrating **JWT**-based authentication, input validation, and secure real-time backend services across all payment and booking endpoints.

PROJECTS

AeroPath AI | Python, FastAPI, React.js, Leaflet, Shapely, A* Algorithm, NOAA API

- Reduced flight path computation time by **40%** by implementing a custom **graph-search algorithm (A*)** with Haversine great-circle heuristics, optimizing pathfinding across large **geospatial datasets** with configurable distance and constraint parameters.
- Enhanced storm-avoidance accuracy by **30%** by integrating **NOAA SIGMET** polygons and applying geospatial collision detection using **Shapely** for real-time airspace boundary validation.
- Built an asynchronous **FastAPI** backend aggregating live **RainViewer** radar data with **sub-150ms** response latency and an avionics-style **React + Leaflet** UI with real-time route overlays.

Credit Card Fraud Detection | Python, NumPy, Pandas, Scikit-learn, Logistic Regression

- Implemented **Logistic Regression from scratch** (gradient descent, sigmoid activation, binary cross-entropy loss) on 284K imbalanced transactions, **4x F1 improvement** over Scikit-learn baseline (F1: 0.11 → 0.45) without using ML libraries.
- Engineered a full **ML preprocessing pipeline** with stratified sampling, mean imputation, and feature scaling (min-max, z-score) improving model convergence across **1000 training epochs** on a 0.2% fraud-rate dataset.